

Parameter Sweep Report: γ , h_a , a_0

Model: Reconstructed Ma et al. (2024) model · inhibitor $i = 10$ · $15 \times 15 \times 15$ grid = 3,375 simulations · horizon $\tau = 400$

Supplementary ranges & fixed values

Decoupling antibiotic dose from lysis strength and steepness

Sweep setup:

- γ = maximum lysis coefficient (1.1–1.4); h_a = antibiotic Hill coefficient (1–5); a_0 = initial antibiotic dose (1–100, log-spaced).
- a_0 is a treatment variable rather than a randomized strain parameter; γ and h_a use the Supplementary Information randomized ranges.
- All other parameters are fixed at the SI base values: $\alpha = 0.95$, $\beta_{\min} = 0.9$, $c = 0.7$, $h_i = 2$, $\kappa_b = 0.35$, $d_a = 0.02$, $d_b = 1$, $\xi = 0.8$, $\phi_{\max} = 1$.
- Initial conditions: $S = R = 0.2$, nutrient = 4, free $Bla = 0$. The strong inhibitor condition $i = 10$ is fixed to isolate antibiotic pharmacodynamics.

1. Endpoint Analysis

1.1 Diversity Analysis

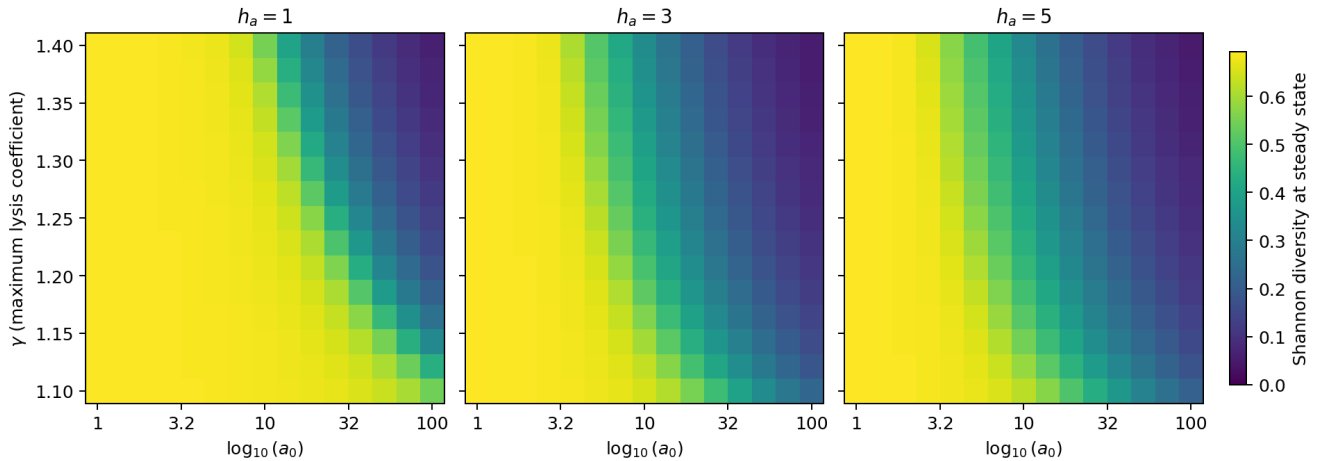


Figure 1. Shannon diversity at steady state. Each panel fixes h_a and shows the $\gamma \times a_0$ response plane.

- Mean H at steady state = 0.455; the theoretical two-population maximum is $\ln 2 \approx 0.693$.
- Diversity is high at low dose because S and R remain relatively balanced, and falls at high dose as S increasingly dominates.
- The strongest gradients run along a_0 , showing that dose is the main driver of community composition in this parameter block.

1.2 Time to reach steady state

- Steady state is defined from the populations: both per-capita growth rates remain below 10^{-3} for the remainder of the run.
- All 3,375 runs reached this criterion by $\tau = 300$; median time = 86.7.

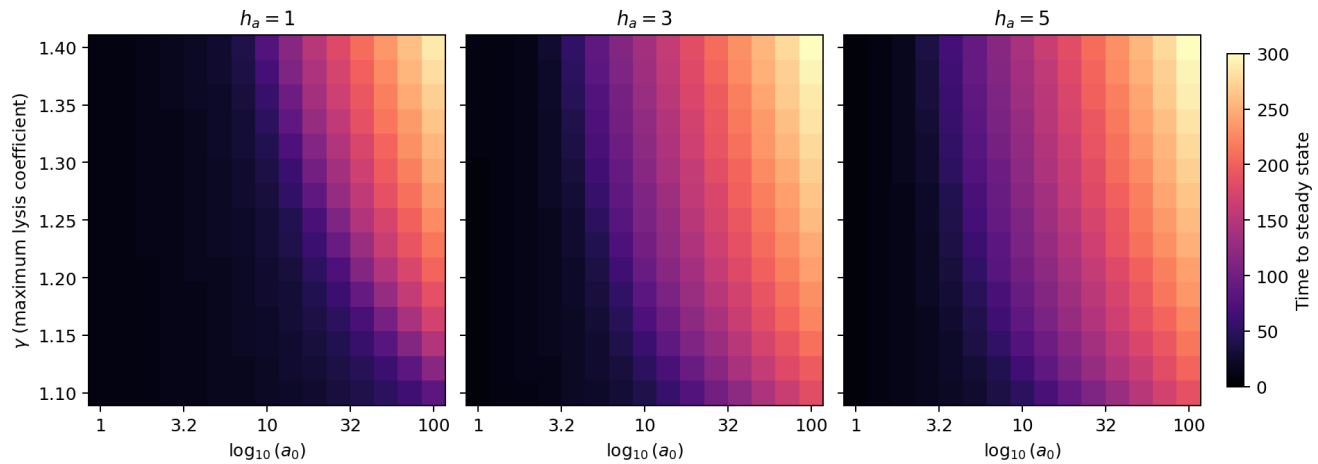


Figure 2. Time to numerical steady state.

Dose-level summary

a_0	Mean final f_R	Mean peak loss	Mean t_{SS}	Mean S recovery
1	0.464	0.0%	8.2	6.7
2.7	0.426	34.3%	18.9	17.0
10	0.214	89.4%	89.0	87.1
27	0.093	98.0%	157.6	155.8
1e+02	0.027	99.9%	246.0	244.0

- Higher dose delays recovery because the community remains in the antibiotic-driven crash phase for longer before regrowing.
- γ changes the maximum lysis intensity and shifts recovery time, while h_a has its largest effect near the transition dose $a \approx 1$.

1.3 Final resistant fraction at steady state

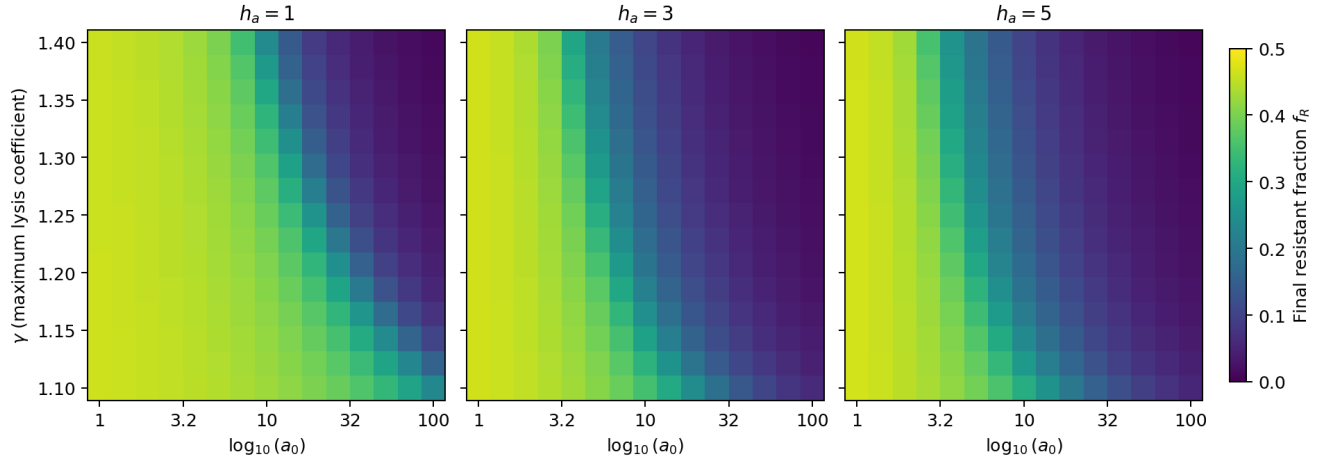


Figure 3. Final resistant fraction $f_R = n_R/(n_S+n_R)$.

- f_R ranges from 0.009 to 0.466; no point ends with R as the majority ($f_R > 0.5$).
- 34.9% of the grid ends below $f_R = 0.10$, but none of the runs eliminate R below the project absolute cutoff $n_R < 10^{-4}$.
- At fixed baseline resistance parameters and strong inhibitor, increasing antibiotic intensity strengthens selection against R rather than reversing its direction.

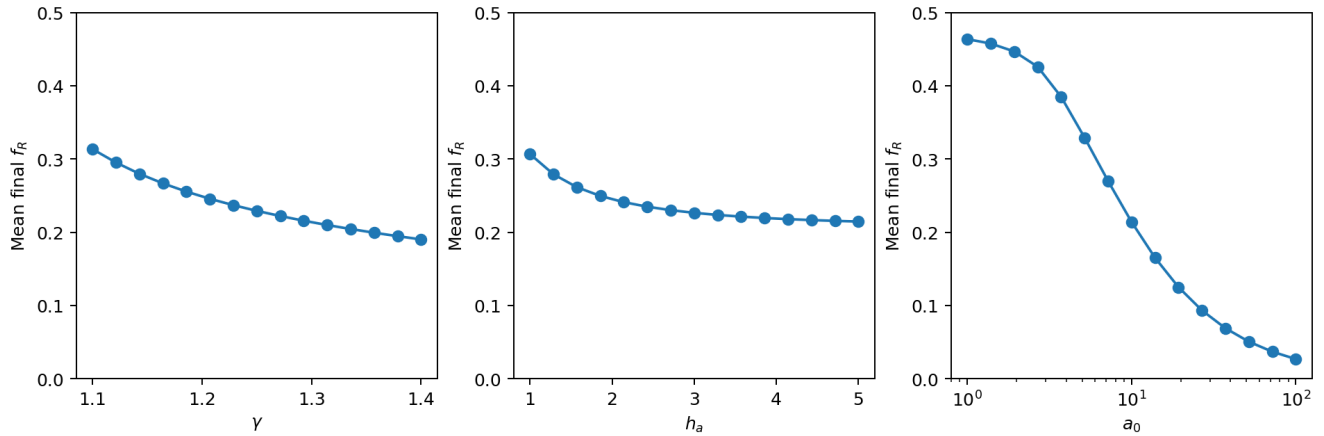


Figure 4. Marginal mean final resistant fraction versus γ , h_a , and a_0 .

- a_0 produces the largest marginal shift in composition; γ is secondary and h_a is the weakest of the three over the specified ranges.

2. Lysis Mechanism and Parameter Decoupling

2.1 How γ , h_a , and dose shape lysis

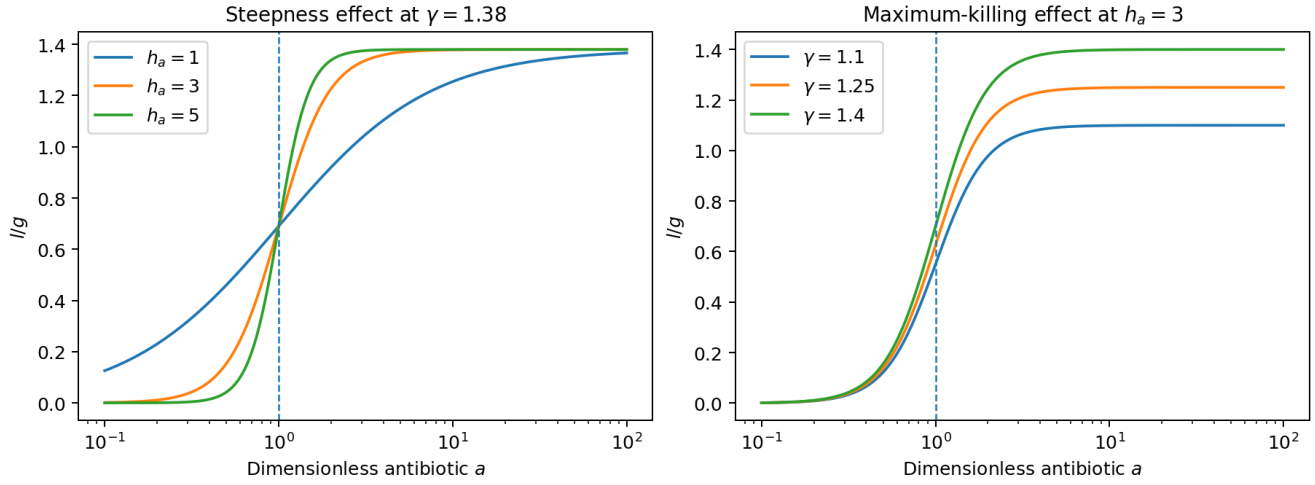


Figure 5. The lysis ratio $I/g = \gamma \cdot a^{h_a} / (1 + a^{h_a})$. Left: h_a changes the transition steepness. Right: γ rescales the saturating maximum.

- At $a = 1$, the Hill term equals 0.5 for every h_a . h_a therefore matters most around the transition region.
- When $a \gg 1$, the Hill term approaches 1 and all curves converge toward $I/g \approx \gamma$; this limits the influence of h_a at high doses.

2.2 Decoupling γ and h_a at the paper dose $a_0 = 2.15$

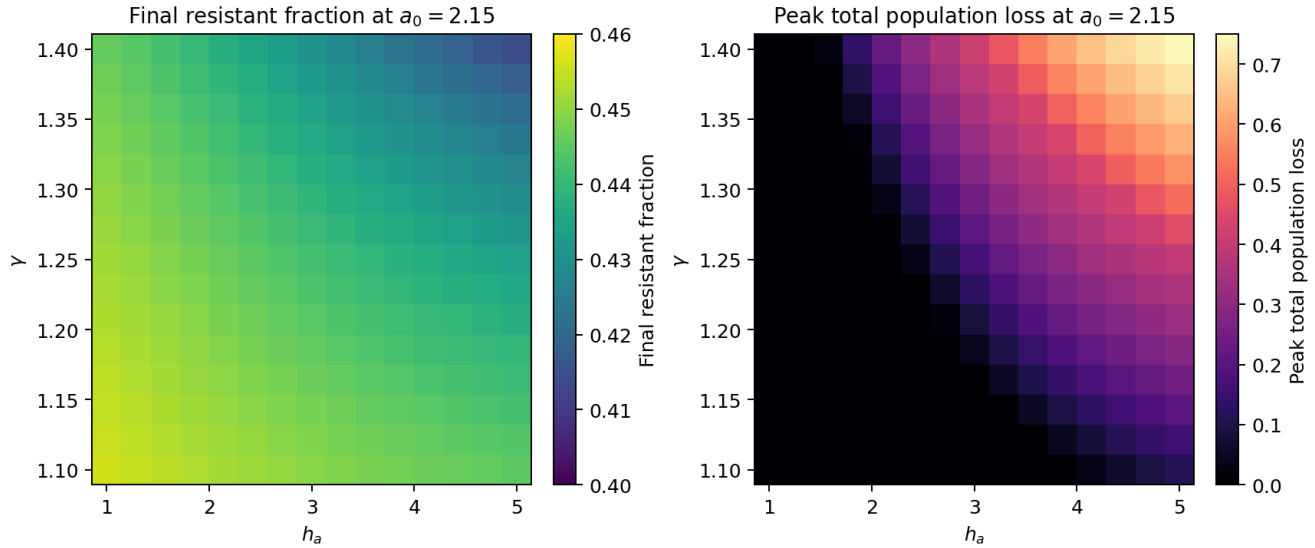


Figure 6. At fixed $a_0 = 2.15$, γ and h_a change the acute crash much more than the final S/R split.

- Across the $\gamma \times h_a$ plane, final f_R spans only 0.414–0.456.
- Over the same plane, peak total population loss spans 0%–74%.

3. Population Loss and Recovery

3.1 Peak total population loss

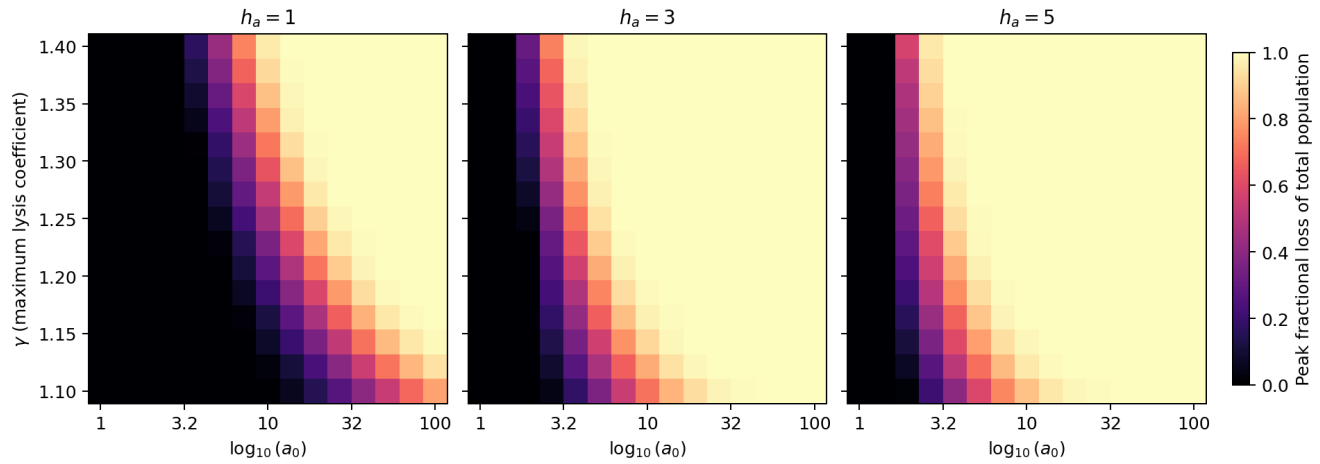


Figure 7. Deepest fractional loss of the total live population during treatment.

- 58.4% of runs lose at least 90% of total population transiently; 51.1% lose at least 99%; 45.6% lose at least 99.9%.
- The deepest bottlenecks occur at high a_0 and stronger lysis. A deep transient crash does not imply final eradication.

3.2 Sensitive-population recovery

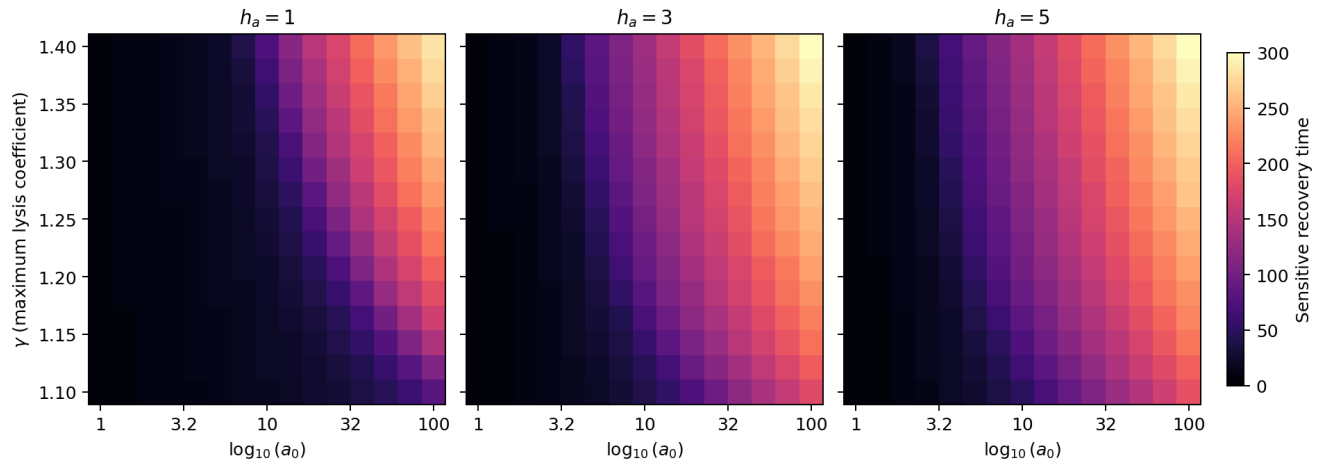


Figure 8. Time for S to return to 90% of its eventual final level after the crash.

- Median S recovery time = 83.3; maximum = 300 model time units.
- Recovery time closely tracks time to steady state because S keeps regrowing after antibiotic activity falls.

3.3 Dose drives both bottleneck depth and composition

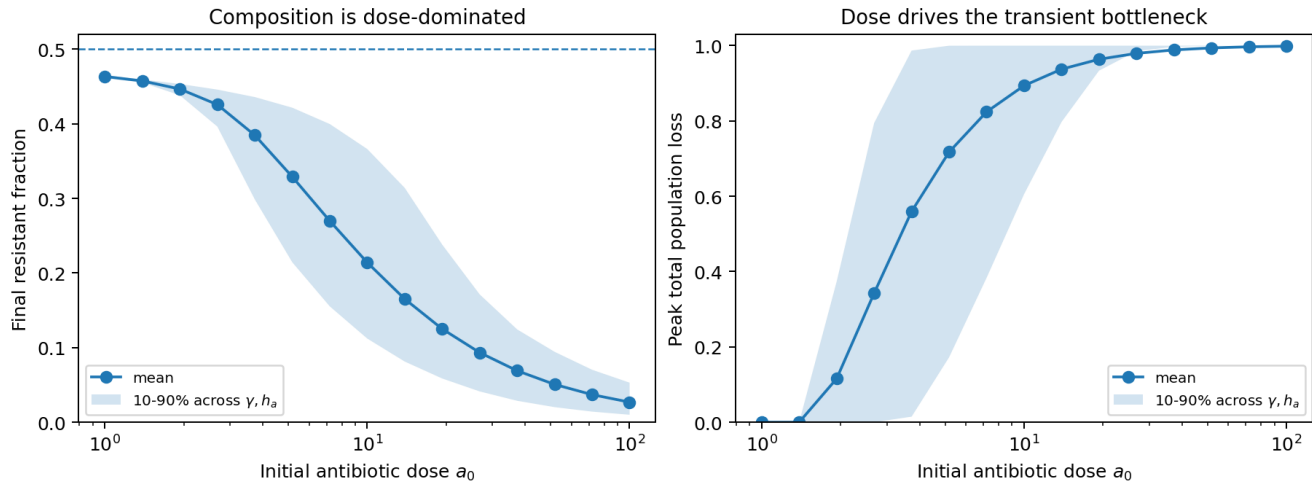


Figure 9. Mean response versus a_0 , with the shaded band showing the 10th-90th percentile across γ and h_a .

- Mean final f_R falls monotonically from ≈ 0.46 at $a_0 = 1$ to ≈ 0.027 at $a_0 = 100$.
- Mean peak total loss rises from 0 at low dose to $\approx 99.9\%$ at $a_0 = 100$.
- Despite these crashes, final total population remains 2.19–4.29, showing substantial regrowth after a single treatment pulse.

Interpretation: the antibiotic parameters strongly control how deep and how long the population crash is. The final resistant-versus-susceptible split is less sensitive at a fixed moderate dose because the same lysis term acts on both populations, with their differential response mediated by β .

4. Representative Dynamics

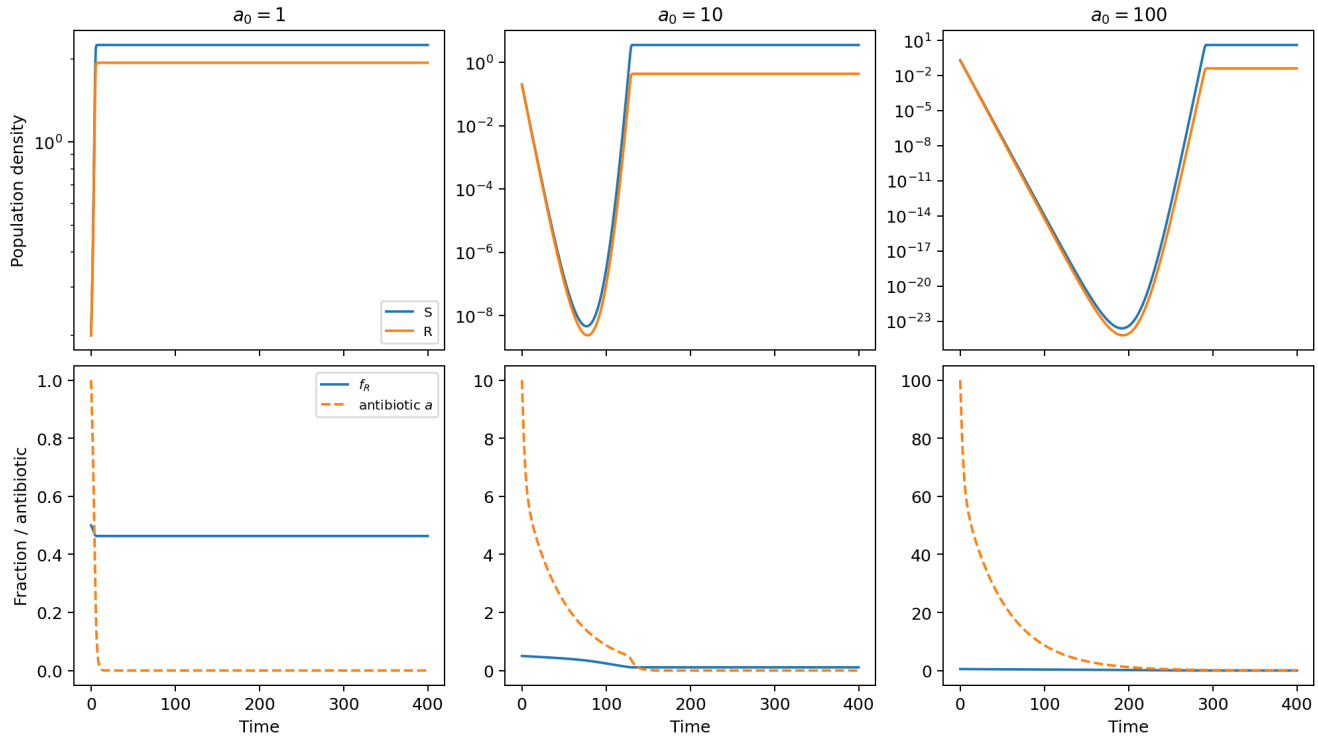


Figure 10. Representative baseline trajectories at $a_0 = 1, 10$, and 100 ($\gamma = 1.38$, $h_a = 3$, $i = 10$). Top: S and R densities on a log scale. Bottom: resistant fraction and antibiotic concentration.

- $a_0 = 1$: little or no bottleneck; S slowly gains because R pays the resistance growth burden.
- $a_0 = 10$: both populations crash, then recover after antibiotic falls; S gains much more strongly during the prolonged dynamics.
- $a_0 = 100$: the model predicts an extremely deep bottleneck followed by regrowth. R ends as a small fraction but is not mathematically eradicated.

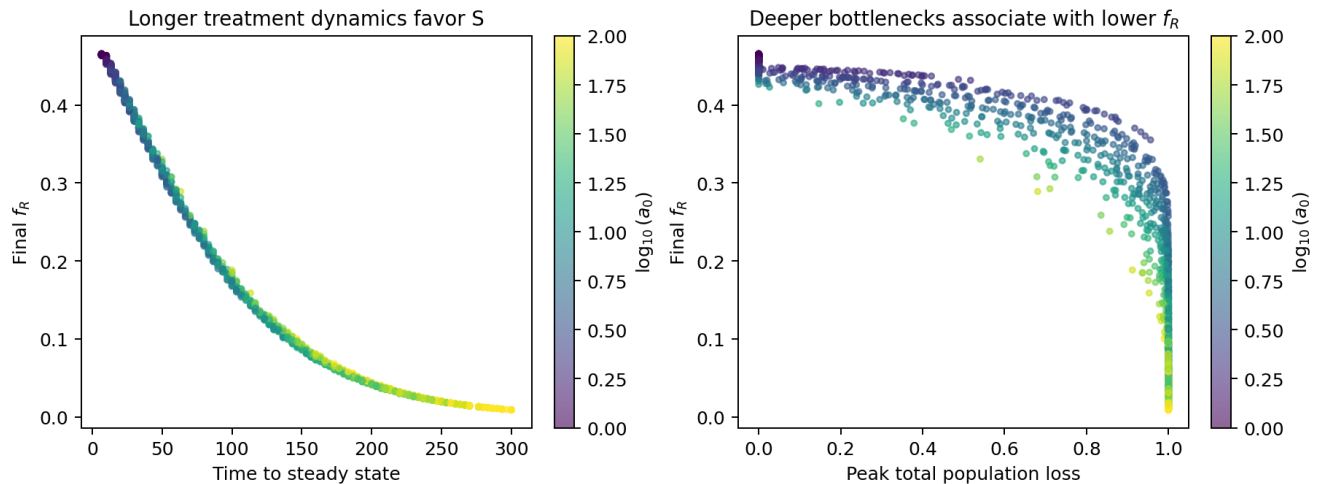


Figure 11. Across the sweep, deeper/longer treatment dynamics associate with lower final f_R . Point color encodes $\log_{10}(a_0)$.

5. Sensitivity Summary and Conclusions

Table 2. Marginal and rank-based sensitivity summaries. These are descriptive summaries of this structured grid, not independent causal coefficients.

Parameter	Range of marginal mean f_R	Spearman vs f_R	Spearman vs peak loss	Spearman vs t_{SS}
γ	0.123	-0.228	0.334	0.217
h_a	0.092	-0.113	0.222	0.104
a_0	0.437	-0.952	0.875	0.954

Main findings:

- Dose a_0 is the dominant parameter in this block. Its Spearman correlation with final f_R is about -0.95 , much stronger than γ or h_a .
- γ and h_a matter for acute pharmacodynamic severity, especially bottleneck depth and recovery time, but have a smaller effect on the final S/R split at the paper's moderate dose.
- At $a_0 = 2.15$, changing γ and h_a moves final f_R by only about 0.042 across their full ranges, while peak total loss moves by about 0.74.
- With $\alpha = 0.95$, $\beta_{\min} = 0.9$, $c = 0.7$ and $i = 10$ fixed, all parameter combinations select against R relative to the initial 50/50 mixture.
- High-dose crash-and-recovery is common: the model can drive populations to extremely low transient density without final extinction, because antibiotic decays/degrades and the populations later regrow.

Modeling caveat: These are dimensionless, deterministic single-dose simulations. A transient density far below one cell in physical units is not biologically meaningful without a stochastic extinction rule and dimensional mapping. High-dose results should therefore be interpreted as “very deep bottleneck” rather than guaranteed clinical eradication.

Reference: Ma HR, Xu HZ, Kim K, Anderson DJ, You L. “Private benefit of β -lactamase dictates selection dynamics of combination antibiotic treatment.” Nature Communications 15, 8337 (2024). Supplementary Information Sections 1–3.